Forecast-Informed Deep Learning for Electricity Price Prediction in the Australian NEM

Group 12
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Abstract

Spot electricity prices in Australia's National Electricity Market (NEM) are highly volatile, non-stationary and prone to extreme price spikes. Using publicly available AEMO market data, we implement and evaluate deep learning architectures for spot price forecasting across the five NEM regions, applying hyperparameter tuning across the model series. We benchmark forecasting performance against AEMO's official predictions to assess whether deep learning approaches can outperform them.

1 Topic

We propose to implement a deep learning-based short-term electricity price forecasting system for the Australian NEM, which comprises five interconnected regional markets (NSW, QLD, SA, TAS, VIC). We will implement and compare a series of models, forecasting spot prices at 5-minute, 30-minute and 60-minute horizons. This will include standard deep learning baselines (MLP, LSTM, Boosting, self-attention/Transformer models) with parameter fine-tuning, creating a collection of models and experimenting with different architectures to determine the best performing model. All models will be trained on identical data splits and evaluated using the same metrics (MAE, RMSE, sMAPE and directional accuracy), with a focus on performance during extreme and negative price events. This methodology was inspired by the benchmarking approach of Gani et al. (2026), which tests whether increased model complexity correlates with improved performance on volatile, regime-switching electricity price series, while relating our findings to the concepts covered in the course. Our objective is to outperform a naive persistence benchmark and AEMO's own published projections whilst using them as model inputs. AEMO releases five-minute pre-dispatch (P5MIN) projections of price, demand and available generation for the coming hour, yet neither reference study uses them, which is where the novelty in our project lies.

2 Significance

Electricity is sold at unregulated spot prices but bought by consumers at regulated rates, so when prices swing wildly, retailers take the hit directly, which can threaten3 Fea both their finances and the stability of the grid (Li et al., 2025). Between October 2021 and January 2022, average NEM spot prices were almost four times higher than the year before and peaked at \$14,000/MWh (Li et al., 2025). More recent data from 2023–2025 shows this hasn't gone away: growing solar uptake now pushes midday prices negative on a large share of intervals, and the evening ramp as solar output drops off is still a major source of extreme price spikes (Gani et al., 2026). Being able to forecast prices accurately in these conditions is useful for retailers managing risk, generators deciding when to bid and system operators trying to keep enough reserve capacity on hand. Gani et al. (2026) found that several recent state-of-the-art time-series models (TimesNet, iTransformer, TimeMixer), despite being top performers on standard, fairly stable benchmark datasets, are often beaten by much simpler recurrent and convolutional models once tested on real NEM data with actual volatility, structural breaks and negative prices. This shows that a bigger or more complex model isn't automatically better; what matters instead is whether a model's assumptions, like explicit spatial structure, sensible loss design, or multi-task regularisation, fit the statistical behaviour of the data it's being used on.

3 Feasibility

Data: We are using the AEMO public dataset, which publishes historical Regional Reference Price (RRP), demand, generation and inter-regional power-flow data at 5-minute resolution for all five NEM regions. We will use `NEMOSIS`, a Python package for querying and pre-processing AEMO NEM data, to pull and clean these same series for all five regions within a data pipeline which all models can use. **(John adds more details here)**.
Acomplished So Far: Over the past four weeks, we have implemented two baseline deep models (MLP regressor and XGBoost regressor) currently configured for 30-minute price forecasting on QLD and NSW. We have also built the full data pipeline that fetches and caches AEMO dispatch price, demand/generation, interconnector, and P5MIN forecast data via NEMOSIS and NEMSEER, constructing a shared feature set (lags, rolling statistics, reserve margins, and AEMO forecast alignment) through a common model interface. This pipeline also handles chronological train/validation/test splitting with a forecast-horizon embargo, ensuring consistent, leakage-free data preparation for all models going forward.

Planned work. We have split into parallel workstreams and organised the implementation into two-week sprints, so each stage exercises concepts covered in lectures. Sprint 1 (Weeks 6–7, 31/08–13/09) covers data retrieval and preprocessing with `NEMSEER`; implementation of an MLP, a Temporal Convolutional Network, a Transformer, an LSTM, and a Gradient Boosting baseline as a non-deep-learning reference point; a shared evaluation pipeline reporting MAE, RMSE, sMAPE, and directional accuracy so every model is scored consistently; and finalising this proposal. This spans course content from data preparation and model selection through to gradient-based learning, convolutional and recurrent architectures, and attention mechanisms. Sprints 2–3 (Weeks 8–9) then consolidate evaluation across the Sprint 1 models and move on to the harder graph-attention multi-task model inspired by Li et al. (2025), applying course content on optimisation and regularisation before building the graph-neural-network layers and the multi-task loss with learned task weights.
Evaluation. We will report MAE, RMSE, sMAPE, and mean directional accuracy on day-ahead and two-day-ahead horizons, with a dedicated breakdown for the upper/lower 5% price tail and negative-price intervals, following Gani et al. (2026), and weighted-F1 for the auxiliary spike-classification task once the multi-task model is implemented, following Li et al. (2025).
Risks and mitigation. The main risks are compute budget for training six model families across five regions and two horizons, mitigated by training on UQ's Rangpur HPC cluster for GPU resources well beyond a personal machine, and the added complexity of the graph-attention model, which is deliberately scheduled for Weeks 8–9 so a working set of simpler baselines and a shared evaluation pipeline already exist first. Given the public data, the `NEMSEER` pipeline, available reference implementations, and the sprint plan's alignment with weekly course content, we consider the project feasible within the semester timeline.

References

Gani, M. O. et al. (2026). "Deep Time-Series Models Meet Volatility: Multi-Horizon Electricity Price Forecasting in the Australian National Electricity Market". In: *arXiv preprint arXiv:2602.01157*.

Li, Y. et al. (Jan. 2025). "Multi-Task Graph Adaptive Learning for Multivariate Electricity Price Short-Term Forecasting in Australia's National Electricity Market". In: *IEEE Transactions on Power Systems* 40.1, pp. 530–541. DOI: 10.1109/TPWRS.2024.3380834.